

NLP: Course Project

The Neural Sentiment Classifier

Due Date: Saturday May 16th, 2026

1. Objective

Design and implement a neural text classifier that maps text reviews to a sentiment probability (0 = Negative, 1 = Positive).

2. Technical Constraints & Requirements

Your team is responsible for meeting the following four system specifications:

Spec A: The Preprocessing Pipeline

- **Input:** Raw strings.
 - **Mechanism:** You must utilize a pre-trained Word2Vec or GloVe model.
 - **Framework:** PyTorch or any one of your choice.
 - **Architecture:** A Feed-Forward Neural Network consisting of:
 1. An input layer matching your embedding dimensions.
 2. At least one hidden layer with a non-linear activation (e.g., ReLU or Tanh).
 3. A final output layer using an activation function that constrains the output to the range $[0, 1]$
 - **Optimization:** You must use a loss function appropriate for **Binary Classification**.
 - **Quantitative Validation:** Report the final accuracy on a held-out test set.
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3. Team Deliverables

One submission per team, containing:

1. **The Engine:** A Python implementation meeting all specs.
2. **The Logic Log:** A brief document (max 1 page) explaining:
 - How the team divided the Specs among the 5 members.

- The dimensionality of your sentence vectors.
 - The results and a brief hypothesis on why the model succeeded or failed.
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3. Evaluation Criteria

Teams will be graded on:

- **Functional Correctness:** Does the code run from start to finish?
 - **Spec Adherence:** Does the model use embeddings and a neural architecture as specified?
 - **Insight:** Does the team understand *why* the model behaves the way it does?
 - **Theory:** What are the theoretical concepts behind the submitted work?
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Best Wishes

Prof. Alaa E. Abdel-Hakim